YUDI KARMA

FULLSTACK DEVELOPER

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PROFESSIONAL SUMMARY

Versatile and results-driven Senior Fullstack Developer with over 8 years of engineering experience architecting and delivering enterprise-scale web platforms, native mobile applications, robust backend microservices, and security authentication solutions. Proven track record consulting Tier-1 financial institutions including Bank Central Asia (BCA), Bank Negara Indonesia (BNI), Bank Tabungan Negara (BTN), CIMB Niaga, Panin Bank, and Bank BJB. Expert in unifying frontend interfaces (React 19/18, TypeScript, Tailwind CSS), backend architectures (Node.js, Express, Java JAX-RS, Laravel 12, PostgreSQL), and native mobile systems (Kotlin, Android SDK, Jetpack Compose, Flutter). Deep domain specialization in banking cryptography (Vasco Digipass, Cronto QR signing), W3C WebAuthn/FIDO2 standards, application shielding (DexGuard RASP, mTLS), real-time VoIP (Avaya WebRTC), and cloud infrastructure (GCP, Dynatrace APM).

TECHNICAL COMPETENCIES

Frontend Web:	React 19, React 18, TypeScript, JavaScript (ES6+), Next.js, Vite, Tailwind CSS, HTML5, CSS3, State Management (Zustand, Redux Toolkit, Context API), Framer Motion, Responsive UI/UX, Recharts / D3.js, WebSockets Client
Backend & Cloud:	Node.js, Express, Java (JAX-RS / RESTful APIs), Laravel 12, PHP 8.3, PostgreSQL, Supabase BaaS, Firebase (Authentication & Firestore), RESTful Microservices, Google Cloud Platform (GCP), Docker, CI/CD Pipelines
Mobile Engineering:	Kotlin, Java, Android SDK/NDK, Jetpack Compose (Material 3), Coroutines, StateFlow, Dagger Hilt, MVVM & Clean Architecture, Room Database, WorkManager, CameraX & Camera 1/2 APIs, EXIF Metadata, Flutter, Dart, React Native
Security & APM:	W3C WebAuthn & FIDO2 Standards, Vasco Digipass Cryptography, Cronto QR Transaction Signing, OneSpan Mobile Security Suite, DexGuard RASP & Dynamic Obfuscation, mTLS Pinning, Avaya Elite WebRTC & SIP, Tetherfi SDK, Dynatrace APM

PROFESSIONAL EXPERIENCE

Senior Fullstack & Mobile Solutions Engineer

Jun 2021 — Present

PT. Phintraco Technology — Enterprise Financial & Security Solutions (Jakarta, Indonesia)

- **Enterprise FIDO2 & WebAuthn Platform:** Architected an end-to-end passwordless biometric authentication platform; engineered Android Credential Manager authenticator client, Java JAX-RS REST Relying Party backend, PostgreSQL database, and modern web administrative dashboard.
- **Phintraco SoftToken Suite & Banking Web Portal:** Developed a dual-channel banking security solution; engineered KarmaMSS native Android app (Jetpack Compose, Vasco Digipass cryptography, Cronto QR scanning) and built a React 18 & TypeScript Internet Banking simulation web portal with real-time OTP validation.
- **KeyBCA Banking Soft Token (Client: Bank Central Asia):** Engineered and maintained KeyBCA mobile client for retail and commercial banking; implemented Vasco Digipass cryptographic token generation algorithms and mutual TLS (mTLS) microservice integrations.
- **Halo BCA & Wondr BNI WebRTC (Clients: BCA & BNI):** Spearheaded Avaya Elite WebRTC and SIP communication integration, authoring Kotlin and Jetpack Compose reference sample apps to handle real-time audio/video streaming and agent routing.
- **Tetherfi Omnichannel Customer Engagement (Clients: Bank BTN & CIMB Niaga):** Integrated Tetherfi SDK into bale by BTN and OCTO Mobile (CIMB Niaga), delivering modular Jetpack Compose UI components for live customer-agent chat and push messaging.
- **Banking Security Consulting (Clients: Panin Bank & Bank BJB):** Consulted client development teams on BJB Digi and Panin Mobile for OneSpan Mobile Security Suite integration, biometric credential binding, and DexGuard RASP application shielding.
- **Dynatrace APM Telemetry:** Configured Dynatrace Real User Monitoring (RUM), transaction latency tracking, and crash alerting across high-volume banking clients.

Full-Stack Mobile & Web Engineer

Nov 2019 — Jun 2021

PT. Derap Kreasi Persada — Enterprise Mobile & Web Systems (Jakarta, Indonesia)

- **Santriqu Fintech (Client: Educational Institutions):** Architected school fintech platform using MVVM Clean Architecture, integrating multiple local payment gateways for automated tuition settlements and digital PPOB transactions.
- **Yipy Operations (Client: Centratama Group Staff):** Developed cross-platform operations utility using Flutter & Dart for Centratama Group facility management staff, featuring real-time ticket dispatching over WebSockets and Google Maps logistics.
- **Warung Babe P2P (Client: Giropos / PT Pos Indonesia Merchant Network):** Engineered native Android P2P gold trading client with Giropos API billing integration and reactive state management for live gold market rate updates.

Native Android Developer

Nov 2018 — Nov 2019

PT. 107, BSD — Native Android Engineering (Tangerang, Indonesia)

- **Snow Remove Camera (Client: Industrial Contractor, Japan):** Developed industrial camera application for snow removal machinery operators in Japan; reverse-engineered low-level Camera 1 & 2 APIs to eliminate hardware kernel panics and injected tamper-proof EXIF GPS/timestamp metadata.
- **Zonapets Social Platform (Client: Hyrax Pet Network):** Built social media application for pet owners featuring bitmap compression pipelines and nested RecyclerView feed caching.

Software Quality Assurance Intern (E-Procurement)

2018 (6 Months)

PT. Mitra Mandiri Informatika — Enterprise Software Systems (Jakarta, Indonesia)

- **Promise Platform (Enterprise E-Procurement):** Authored manual test cases and automated regression suites on Jira for enterprise procurement workflows, verifying transaction integrity across supplier-buyer portals.

KEY FULLSTACK PROJECTS & ARCHITECTURAL CASE STUDIES

1. Enterprise FIDO2 & WebAuthn Biometric Platform — PT. Phintraco Technology

Project Overview: Engineered an end-to-end passwordless biometric authentication platform adhering strictly to W3C WebAuthn and FIDO2 standards. The system eliminates credential stuffing and phishing attacks by leveraging public-key cryptography and hardware-backed authenticators, linking an Android client, a high-throughput Java REST Relying Party backend, and an administrative console for managing multi-tenant credentials and challenge logs.

- **Web Tech:** Modern Web Portal, Tailwind CSS, WebAuthn Browser API, Fetch Client, Responsive Console
- **Backend Tech:** Java (JAX-RS), PostgreSQL, W3C WebAuthn Relying Party Specification, Digital Asset Links
- **Mobile Tech:** Kotlin, Jetpack Compose, Android Credential Manager API, Play Services FIDO2 Authenticator

2. Phintraco SoftToken Suite & Internet Banking Portal — PT. Phintraco Technology

Project Overview: Architected a comprehensive dual-channel banking security showcase that connects a high-fidelity React 18 Internet Banking web portal with a native Android SoftToken client (KarmaMSS). Provides seamless out-of-band transaction signing using high-density visual Cronto QR cryptograms, offline cryptographic credential activation, and real-time counter-based OTP verification against banking audit ledgers.

- **Web Tech:** React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons, Dynamic Cronto QR Matrix Generator
- **Backend Tech:** Node.js / REST APIs, Vasco Digipass OTP Verification Microservice, Audit Trail Ledger
- **Mobile Tech:** Kotlin, Jetpack Compose (Material 3), Vasco Digipass Cryptography, Camera Cronto QR Scanner, Biometrics

3. LogamHub Enterprise Gold Inventory & SaaS Platform — Private Client (Irvan Gold Roger)

Project Overview: Architected a distributed gold trading and inventory auditing ecosystem designed for retail jewelry networks. Features an offline-first mobile client for in-store staff with automated background sync, live commodity price feeds, camera-based barcode auditing, and a centralized web dashboard for multi-store stock reconciliation.

- **Web Tech:** React 18, TypeScript, Tailwind CSS, Interactive Price Analytics Charts, Responsive Admin UI
- **Backend Tech:** Supabase Cloud BaaS (PostgreSQL, Row Level Security, Realtime Subscriptions), Node.js Services
- **Mobile Tech:** Kotlin, Jetpack Compose, Room Database (Offline Cache), WorkManager Sync, CameraX ML Kit Scanning

4. FinTrack AI Personal Finance & Receipt Intelligence — Independent Product

Project Overview: Developed a full-stack personal finance application with an automated multimodal receipt extraction pipeline. Users upload raw paper receipt photos which are parsed via Google Gemini 1.5 Flash Vision API into structured merchant, tax, and itemized transaction data, with dynamic budget visualization and real-time ledger synchronization.

- **Web Tech:** React 19, TypeScript, Tailwind CSS, Recharts Visualization, Lucide Icons, Drag & Drop Upload UI
- **Backend Tech:** Node.js, Express, Firebase Firestore & Auth, Google Gemini 1.5 Flash Vision Multimodal OCR API
- **Mobile Tech:** Progressive Web App (PWA) / Responsive Mobile Viewport, Camera Hardware Capture Integration

5. Warkop Modern POS & Multi-Tenant F&B Engine — Independent SaaS

Project Overview: Engineered a multi-tenant Point-of-Sale (POS) and self-ordering platform for food and beverage operations. Implements database tenant isolation, role-based access control (Admin, Cashier, Kitchen Barista), encrypted table-specific QR ordering, and automated end-of-day sales revenue settlement dashboards.

- **Web Tech:** Responsive Web Interface, Blade UI, Tailwind CSS, Dynamic Encrypted QR Code Generator
- **Backend Tech:** Laravel 12, PHP 8.3, PostgreSQL, Eloquent ORM, Role-Based Access Control (RBAC), Queue Workers
- **Mobile Tech:** Mobile-Optimized Self-Order Web App for Customer Touchscreens & Handheld POS Devices

6. Wedly Finance Editorial SaaS Platform — Independent Product

Project Overview: Designed an editorial wedding financial management web platform supporting dual-role access for couples and planners. Features real-time expense forecasting, interactive milestone timelines, fluid micro-interactions, and a zero-dependency vendor budget comparison matrix.

- **Web Tech:** React 18, TypeScript, Tailwind CSS, Framer Motion, Responsive Editorial UI
- **Backend Tech:** Node.js, RESTful Endpoints, LocalStorage Caching & Cloud Synchronization
- **Mobile Tech:** Touch-Optimized Responsive Mobile Web Application

7. Enterprise WebRTC & Real-Time Omnichannel Suite — PT. Phintraco Technology

Project Overview: Engineered reference communication architecture for enterprise banking clients (BCA, BNI, Bank BTN, CIMB Niaga). Demonstrates end-to-end integration of WebRTC real-time voice/video calls, SIP signaling, and omnichannel live agent routing into production mobile and web environments.

- **Web Tech:** Avaya WebRTC Web Portal, WebSockets Signaling, HTML5 Media Streams, Responsive Agent Dashboard
- **Backend Tech:** Avaya Oceana & Elite WebRTC Media Gateway, SIP PBX Server, Session Border Controller (SBC)
- **Mobile Tech:** Kotlin, Jetpack Compose, Avaya WebRTC SDK, Tetherfi SDK, Audio/Video Stream Lifecycle Tuning

LICENSES & PROFESSIONAL CERTIFICATIONS

- **Google Cloud Certified Associate Cloud Engineer** — Google Cloud (Valid: May 2025 — May 2028)
- **Google Cloud Certified Cloud Digital Leader** — Google Cloud (Valid: Mar 2025 — Mar 2028)
- **Dynatrace Certified Associate** — Dynatrace (Valid: Mar 2026 — Mar 2028)
- **Dynatrace Certified Essentials** — Dynatrace (Valid: Mar 2026 — Mar 2028)
- **Avaya Infinity™ Design and Workflows** — Avaya (Valid: Sep 2025 — Lifetime)
- **Google Developer Kejar (Android Developer Specialist)** — Google x Dicoding (2019)
- **Technical Support Certification** — Balai Pelatihan Kerja (2018)

EDUCATION

Bachelor of Applied Science in Informatic Engineering (GPA: 3.56 / 4.00) **2014 — 2018**
Politeknik Aceh — Banda Aceh, Indonesia